

The Evolution of Tuberculosis Community Transmission Science and Its Ethical and Legal Constructs

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Tuberculosis remains the leading infectious diseases killer globally. Although tuberculosis incidence in the United States is low by global standards, reported tuberculosis cases and incidence have increased annually since 2020 and contributes to substantial morbidity and mortality rates. Public health strategies to reduce or eliminate tuberculosis requires earlier tuberculosis disease diagnosis and treatment, and targeted testing and treatment of tuberculosis infection among contacts and high-risk groups to prevent future tuberculosis disease. Adjunctive postdiagnosis infection control interventions seek to prevent airborne spread from one person to another during the period when a person with tuberculosis is considered most infectious through often prolonged and/or mandated individual isolation. Such practices are complicated, however, by a lack of clinically reliable predictors of individual infectiousness, leading to uncertainty in the optimal duration of isolation

restrictions. Transmission risk is further modified by environmental conditions, duration and proximity of exposures, and other factors. Balancing the community benefits derived through public health infection control interventions with the possible harms of prolonged isolation restrictions remains ethically, legally, and clinically challenging.

In April 2024, *Clinical Infectious Diseases (CID)* published the National Tuberculosis Coalition of America (NTCA) guidelines on respiratory isolation and restrictions for reducing pulmonary tuberculosis transmission in community settings [1]. These guidelines fill an important gap. Surprisingly, no such tuberculosis infection control recommendations for community settings previously existed from national or federal agencies such as the Centers for Disease Control and Prevention. This novel and contemporary set of recommendations provides needed guidance on public health practices related to respiratory isolation and restrictions for persons with pulmonary tuberculosis in community settings (ie, nonhealthcare and noncongregate settings). Notably, they provide an evidence-based framework to estimate infectiousness for the individual with tuberculosis before and, in particular, early in the course of antituberculosis therapy. A key message of the guidelines is that net tuberculosis transmission risk must be considered in the clinical, microbiologic and social/environmental context of each individual,

given highly variable individual circumstances. Implementation of NTCA guidance would shift the paradigm from a one-size-fits-all approach to isolation restrictions to a more nuanced and tailored one that seeks to maximize community benefits while mitigating harms for individuals with tuberculosis.

In this issue of *The Journal of Infectious Diseases (JID)* we extend the reach and impact of these new guidelines by providing 4 additional articles that further highlight important aspects on the individual transmission risk of tuberculosis and the framework for considering the type and duration of isolation of the person with tuberculosis. First, Karakousis and Mooney [2] discuss the history of respiratory isolation for tuberculosis including that the recognition of tuberculosis as an infection transmitted by airborne exposure is only relatively recent. Second, Nathavitharana et al [3] provide data on which measures of bacillary burden are or are not predictive of tuberculosis transmission risk and, importantly, support a key message that effective antituberculosis therapy rapidly decreases the risk of tuberculosis transmission. Third, Kates and Shah [4] present the novel and complex process for integrating bioethics into public health guideline development, both lessons from tuberculosis and guidance for the future. Finally, in their article, Bonomo and Shah [5] assess and explore the legal obligations in developing

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isolation policies for persons with tuberculosis. They emphasize the critical need to balance respect for individual rights and provision of the public health goal to limit the risk of tuberculosis transmission from one individual to another.

These engaging and informative articles being published in *JID* extend the reach and depth of the NTCA guidelines in *CID*. At *JID*, we are excited by the opportunity to work collaboratively with *CID* and across the Infectious Disease Society of America journals to bring these complementary discussions and data to our readers.

Note

Potential conflicts of interest. All authors: No reported conflicts.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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